

Applied Mathematics for Chemists

(화학수학)

Lecture 10

Differential Equations



Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

Terminology

- Order of DE (계수): order of the highest derivative
- Linear (선형) vs. nonlinear (비선형)
 - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Nonlinear: otherwise

$$(1 - y) \frac{dy}{dx} + 2y = e^x, \quad \frac{d^2 y}{dx^2} + \sin y = 0, \quad \dots$$

Linear Differential Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

General and Particular Solutions

(일반해와 특수해)

- **Solution:** any function that reduces the DE to an identity when substituted into the DE

For a DE of order n ,

- **General solution:** contains n arbitrary constants
- **Particular solution:** contains no arbitrary constants
 - Particular solutions can be generated from the general solution



Additional Conditions

- For a DE of order n , if n additional conditions are given, one can determine the arbitrary constants of the general solution
- Initial conditions (IC, 초기 조건): values of the function and its first $(n - 1)$ derivatives at a single point
- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

Solving a Differential Equation

1. Obtain the **general solution**: hardest part!
 2. If initial/boundary conditions are given, substitute the conditions to **determine the coefficients**
- No universal method to solve all DEs
 - For specific forms of DEs, we have general solving methods

First-Order ODEs

$$F\left(y, \frac{dy}{dx}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

Separable Variables

- Separable DE:

$$g(y) \frac{dy}{dx} = f(x)$$

This DE can be manipulated into

$$g(y) dy = f(x) dx$$

$$\int g(y) dy = \int f(x) dx + C$$

This equation provides the **general solution**



Exact Differential Equations

- Pfaffian form:

$$M(x, y) \, dx + N(x, y) \, dy = 0$$

If the differential equation is **exact**,
there is a function $f(x, y)$ such that

$$df = M(x, y) \, dx + N(x, y) \, dy = 0$$

which implies that $f(x, y) = k$ (constant)

How to Find $f(x, y)$

$$df = M(x, y) dx + N(x, y) dy = 0$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_C df$$

where C is a curve connecting (x_0, y_0) and (x_1, y_1) . Take a rectangular path from (x_0, y_0) to (x_1, y_0) to (x_1, y_1) :

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0) dx + \int_{y_0}^{y_1} N(x_1, y) dy$$

This gives the relationship between x and y .



Integrating Factors

- Pfaffian form:

$$M(x, y) dx + N(x, y) dy = 0$$

If the differential equation is **inexact**, but we know the **integration factor** $g(x, y)$ that makes

$$g(x, y)M(x, y) dx + g(x, y)N(x, y) dy = 0$$

an exact differential equation, we can solve this DE.



Linear Equations

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{dy}{dx} + P(x)y = f(x)$$

- General solution:

$$y = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx + C e^{-\int P(x)dx}$$

Today's Class

- Higher-order ODEs: homogeneous and linear
 - Superposition principle
 - Reduction of order
- Homogeneous linear ODEs with constant coefficients
 - Trial solution and characteristic equation
 - Application: harmonic oscillator



Higher-Order ODEs: Linear Case

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
 - Homogeneous (§ 12.2)
 - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs



Homogeneous Linear Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If $y_1(x), y_2(x), \dots, y_k(x)$ are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants $c_1, c_2, \dots, c_k$.



Example

$y_1 = \cos x$ and $y_2 = \sin x$ are the solutions of the differential equation

$$\frac{d^2y}{dx^2} + y = 0.$$

Show that a linear combination of y_1 and y_2 is also its solution.

For arbitrary constants c_1 and c_2 , the linear combination is

$$y = c_1 y_1 + c_2 y_2 = c_1 \cos x + c_2 \sin x$$

Substitution leads to

$$\begin{aligned} \frac{d^2y}{dx^2} + y &= \frac{d^2}{dx^2} (c_1 \cos x + c_2 \sin x) + (c_1 \cos x + c_2 \sin x) \\ &= -c_1 \cos x - c_2 \sin x + (c_1 \cos x + c_2 \sin x) = 0 \end{aligned}$$

Homogeneous Linear Second-Order ODEs

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

If a solution y_1 is known,
we can obtain **the other solution y_2**
by following a systematic procedure
→ **reduction of order** (계수내림)

Reduction of Order

- Standard form

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0$$

- With known $y_1(x)$, define $y(x) = u(x)y_1(x)$:

$$\frac{dy}{dx} = u \frac{dy_1}{dx} + y_1 \frac{du}{dx}$$

$$\frac{d^2 y}{dx^2} = u \frac{d^2 y_1}{dx^2} + 2 \frac{du}{dx} \frac{dy_1}{dx} + y_1 \frac{d^2 u}{dx^2}$$

Substitution!

Rearrangement

$$\frac{d^2y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y$$

$$= u \left(\frac{d^2y_1}{dx^2} + P \frac{dy_1}{dx} + Qy_1 \right) + y_1 \frac{d^2u}{dx^2} + \left(2 \frac{dy_1}{dx} + Py_1 \right) \frac{du}{dx}$$

$= 0$

$$= y_1 \frac{d^2u}{dx^2} + \left(2 \frac{dy_1}{dx} + Py_1 \right) \frac{du}{dx}$$

$$= 0$$

“Reduction of Order”

$$y_1 \frac{d^2 u}{dx^2} + \left(2 \frac{dy_1}{dx} + P y_1 \right) \frac{du}{dx} = 0$$

Let $w = du/dx$;

$$y_1 \frac{dw}{dx} + \left(2 \frac{dy_1}{dx} + P y_1 \right) w = 0$$

(First-order DE)

Solving First-Order ODE

$$y_1 \frac{dw}{dx} + \left(2 \frac{dy_1}{dx} + P y_1 \right) w = 0$$

Separation of variables:

$$\frac{dw}{w} = - \frac{2(dy_1/dx)}{y_1} dx - P dx$$

Integration:

$$\ln w = - \ln y_1^2 - \int P dx + C$$

$$w y_1^2 = C' e^{- \int P dx}$$

Going Back to $u(x)$

$$w y_1^2 = C' e^{-\int P dx}$$

$$w = \frac{C'}{y_1^2} e^{-\int P dx}$$

Since $w = du/dx$,

$$\frac{du}{dx} = \frac{C'}{y_1^2} e^{-\int P dx}$$

$$u = \int \frac{C'}{y_1^2} e^{-\int P dx} dx + C''$$

Another Solution

$$u = \int \frac{C'}{y_1^2} e^{-\int P dx} dx + C''$$

For $y(x) = u(x)y_1(x)$, choose $C' = 1$ and $C'' = 0$:

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

Example

The function $y_1 = x^2$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0.$$

Find the general solution.

The standard form of the equation is

$$\frac{d^2 y}{dx^2} - \frac{3}{x} \frac{dy}{dx} + \frac{4}{x^2} y = 0$$

and $P(x) = -3/x$. Now use

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

Example

The function $y_1 = x^2$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0.$$

Find the general solution.

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx, \quad P(x) = -\frac{3}{x}$$

Substitution leads to

$$y_2(x) = x^2 \int \frac{1}{x^4} e^{\int \frac{3}{x} dx} dx = x^2 \int \frac{1}{x^4} e^{3 \ln x} dx = x^2 \int \frac{1}{x^4} x^3 dx = x^2 \int \frac{1}{x} dx$$

$$y_2(x) = x^2 \ln x$$

Example

The function $y_1 = x^2$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0.$$

Find the general solution.

$$y_2(x) = x^2 \ln x$$

The general solution is a linear combination of $y_1(x)$ and $y_2(x)$:

$$y(x) = c_1 y_1(x) + c_2 y_2(x) = c_1 x^2 + c_2 x^2 \ln x$$

Linear ODEs with Constant Coefficients

- Homogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Inhomogeneous

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

Homogeneous Case

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Trial solution (시험해): $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

Roots of Characteristic Equation

$$a\lambda^2 + b\lambda + c = 0$$

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

1. Two roots are real and distinct ($b^2 - 4ac > 0$)
2. Two roots are real and equal ($b^2 - 4ac = 0$)
3. Two roots are complex numbers ($b^2 - 4ac < 0$)

(1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

Example 12.2

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0.$$

The characteristic equation is

$$\lambda^2 + \lambda - 2 = 0$$

whose solutions are

$$\lambda_1 = 1, \lambda_2 = -2$$

Hence, the general solution is

$$y = c_1 e^x + c_2 e^{-2x}$$

(2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution: $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} dx = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0.$$

The characteristic equation is

$$\lambda^2 - 10\lambda + 25 = 0$$

whose solution is

$$\lambda = 5$$

Hence, the general solution is

$$y = c_1 e^{5x} + c_2 x e^{5x}$$

(3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$y = c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x}$$

(3) Conjugate Complex Roots

$$y = c_1 e^{\alpha x} e^{i\beta x} + c_2 e^{\alpha x} e^{-i\beta x}$$

Since $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$,

$$\begin{aligned} y &= c_1 e^{\alpha x} (\cos \beta x + i \sin \beta x) + c_2 e^{\alpha x} (\cos \beta x - i \sin \beta x) \\ &= e^{\alpha x} \left[\underbrace{(c_1 + c_2)}_{c'_1} \cos \beta x + i \underbrace{(c_1 - c_2)}_{c'_2} \sin \beta x \right] \end{aligned}$$

$$y = e^{\alpha x} (c'_1 \cos \beta x + c'_2 \sin \beta x)$$

Example

Find the general solution to the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 7y = 0.$$

The characteristic equation is

$$\lambda^2 + 4\lambda + 7 = 0$$

whose solutions are

$$\lambda_1 = -2 + i\sqrt{3}, \lambda_2 = -2 - i\sqrt{3}$$

Hence, the general solution is

$$y = e^{-2x} [c_1 \cos(\sqrt{3}x) + c_2 \sin(\sqrt{3}x)]$$